Conservation of Momentum

Answer key

Conventions used throughout

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Kinds of collision (from the lesson):** elastic (kinetic energy is conserved), totally inelastic (the objects stick together), superelastic (kinetic energy increases). Momentum is conserved in all of them when the net outside force is zero.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

Predicting four collisions

Let's try to predict the results of some collisions using our knowledge of momentum. Two carts of equal mass (m) collide.

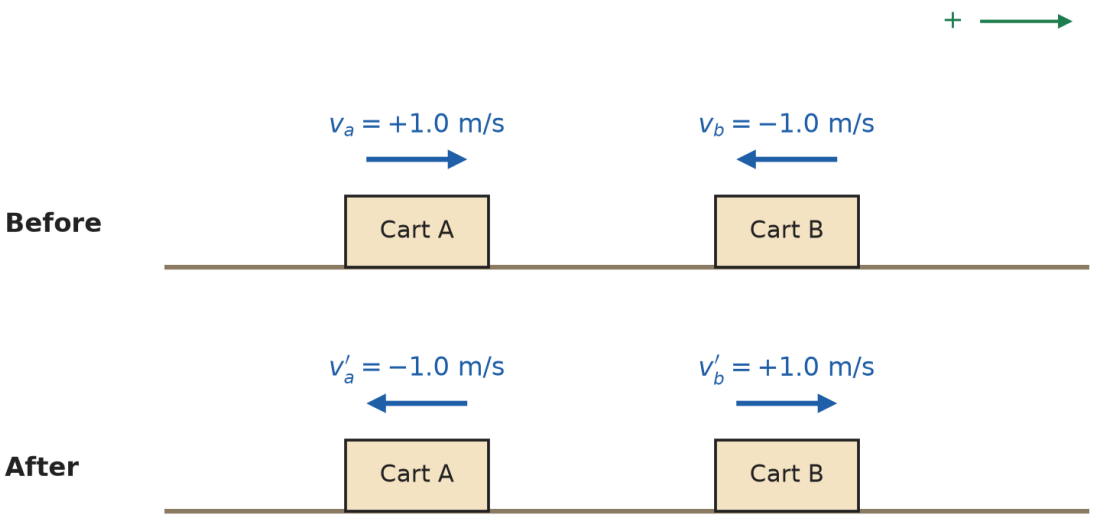
For each case below:

1. Sketch your prediction for the results of the collision. Use arrows to indicate the direction and magnitude of the carts' velocities after the collision. Note that we use the *prime* symbol (i.e. v_2') to show quantities *after* the collision.
2. Observe the actual collision using the dynamics carts set-up in the room. Ignore small changes in velocity due to friction and vibrations. Do your predictions agree? If not, draw a second set of vectors in your sketch.
3. In the middle box, compare the magnitude of the velocities (<, >, =) before and after the collision..
4. Determine the total momentum of the entire system before and after the collision. To keep the numbers simple, assume the **mass of the cart is 0.5 kg and the initial velocities are 1.0 m/s**.

Take + to the right. Cart A starts on the left moving right, so $v_a = +1.0$ m/s. Each cart has m = 0.5 kg. The total momentum is $p_{total} = mv_a + mv_b$, with the signs of the velocities.

Case 1 The Bouncy Collision A. The two carts collide using repelling magnets. $v_a = -v_b$ This is an example of an *elastic* collision (kinetic energy is conserved since there is no source of energy loss).

Case 1: before and after



The carts are identical and meet with equal speeds, so by symmetry each one bounces back with the speed it came in with. That keeps both the momentum and the kinetic energy the same.

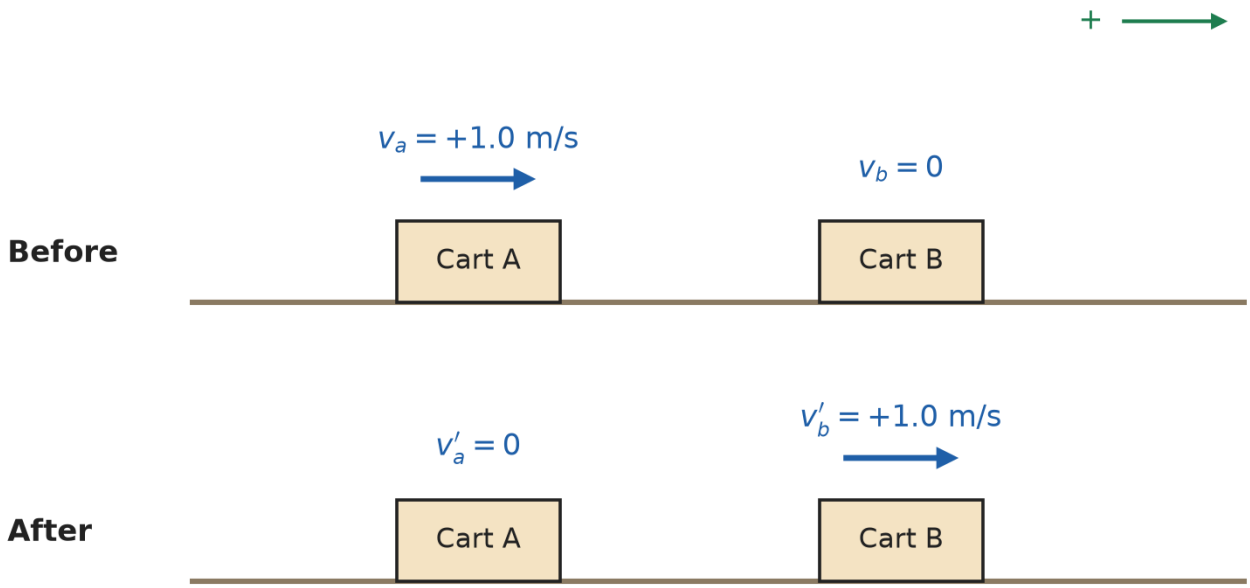
Speeds: $|v_a| = |v'_a| \qquad |v_b| = |v'_b|$

Before: $p_{total} = (0.5)(+1.0) + (0.5)(-1.0) = 0 \text{ kg·m/s}$

After: $p'_{total} = (0.5)(-1.0) + (0.5)(+1.0) = 0 \text{ kg·m/s}$

Case 2 The Bouncy Collision B. The two carts collide using repelling magnets. $v_b = 0$. This is an example of an *elastic* collision.

Case 2: before and after



With equal masses in an elastic collision, the moving cart stops and the cart that was at rest leaves with the same velocity: A hands all of its momentum and kinetic energy to B.

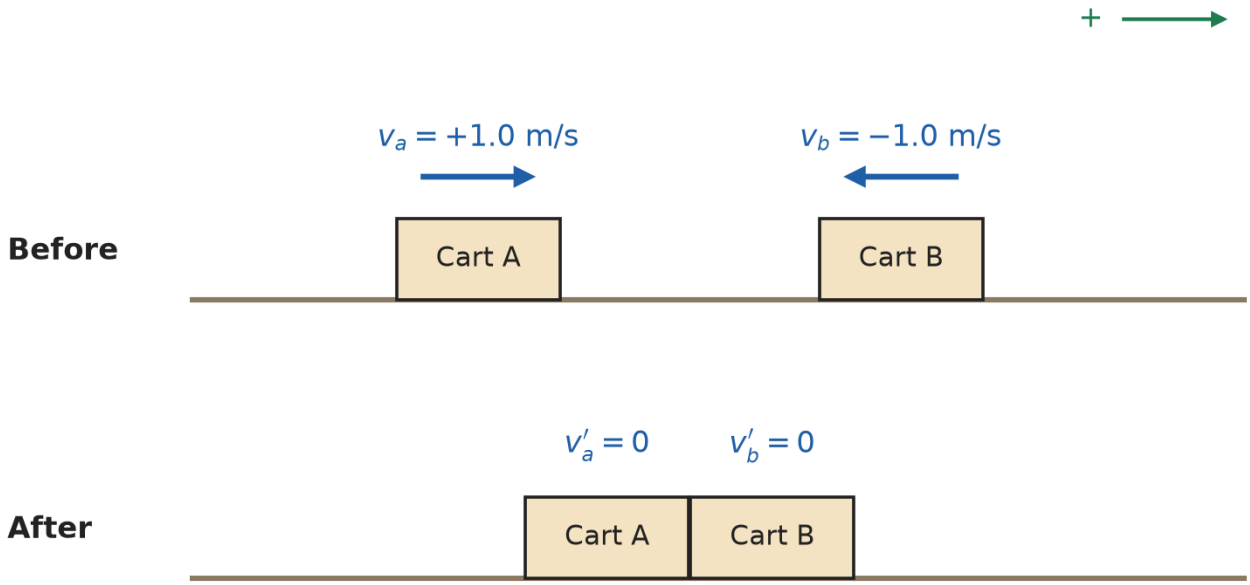
Speeds: $|v_a| = |v'_b|$

Before: $p_{total} = (0.5)(+1.0) + (0.5)(0) = +0.5 \text{ kg·m/s}$

After: $p'_{total} = (0.5)(0) + (0.5)(+1.0) = +0.5 \text{ kg·m/s}$

Case 3 The Sticky Collision! The two carts collide using velcro. $v_a = -v_b$ This is an example of an *totally inelastic* collision (when two objects stick together so kinetic energy is NOT conserved).

Case 3: before and after



The carts stick, so they share one final velocity. The total momentum before is zero, so the joined carts must be at rest. All of the kinetic energy is lost.

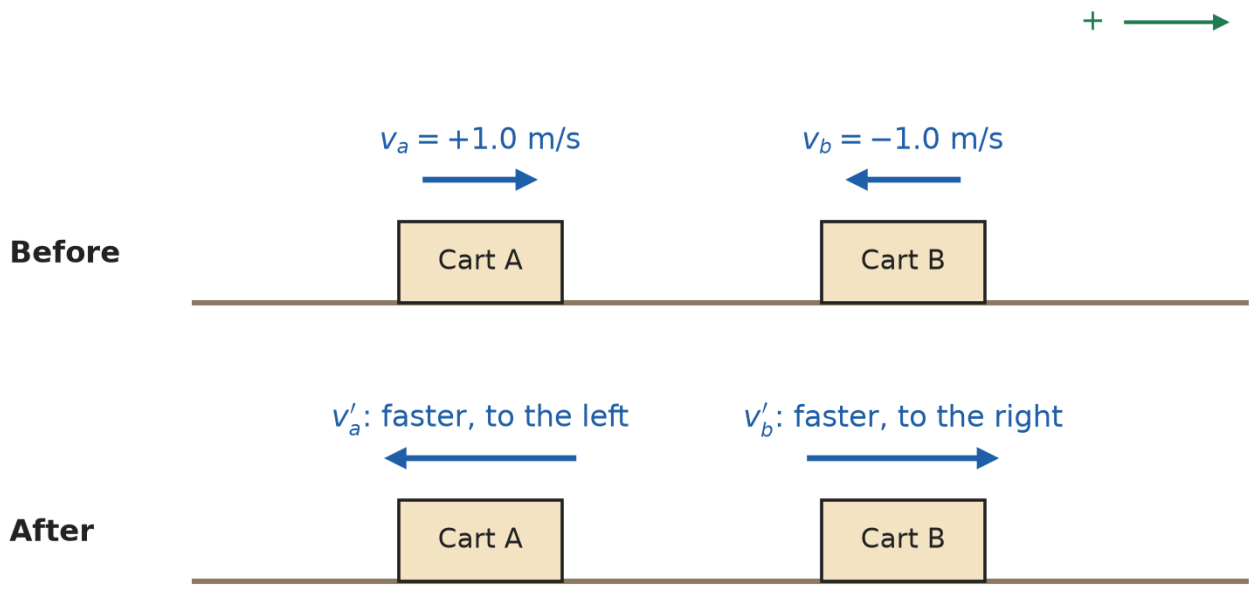
Speeds: $|v_a| > |v'_a| \qquad |v_b| > |v'_b|$

Before: $p_{total} = (0.5)(+1.0) + (0.5)(-1.0) = 0 \text{ kg·m/s}$

After: $p'_{total} = (0.5 + 0.5)(0) = 0 \text{ kg·m/s}$

Case 4 The Explosion! The two carts collide with loaded springs. $v_a = -v_b$ This is an example of a *superelastic* collision (kinetic energy increases).

Case 4: before and after



The springs add kinetic energy, so both carts leave faster than they arrived. The total momentum must still be zero, so the two carts leave with equal speeds in opposite directions. (How much faster depends on the springs, so no number can be given.)

Speeds: $|v_a| < |v'_a| \qquad |v_b| < |v'_b|$

Before: $p_{total} = (0.5)(+1.0) + (0.5)(-1.0) = 0$ kg·m/s

After: $p'_{total} = (0.5)(-v') + (0.5)(+v') = 0$ kg·m/s

FINAL ANSWER

Case 1: carts swap directions, same speeds; $p_{total} = 0$ before and after

Case 2: A stops, B leaves at +1.0 m/s; $p_{total} = +0.5$ kg·m/s before and after

Case 3: both carts stop; $p_{total} = 0$ before and after

Case 4: both leave faster, in opposite directions; $p_{total} = 0$ before and after

TYPES OF COLLISIONS

Is momentum conserved?

Does **momentum** appear to be conserved in each case? Is the final total momentum the same as the initial total momentum of the two-cart system?

Yes. In all four cases the final total momentum equals the initial total momentum (0, +0.5, 0 and 0 kg·m/s). Kinetic energy is different: it stays the same in cases 1 and 2, is lost in case 3 and increases in case 4. Momentum is conserved whether or not kinetic energy is.

FINAL ANSWER

Yes: the total momentum is the same before and after in every case

SIMULATION

PhET Collision Lab

Numbers not given: the momenta in 1a, 1b, 2a and 2b come from your own run of the simulation (they depend on the masses and velocities it starts with), so the key cannot give them. It gives what the totals must show.

1. Before clicking on “play”, record the initial momentum of each object.

   a) Add the two initial momenta (momenta is the plural of momentum). Remember, these are vectors that have directions so add accordingly. Record your answer below.

   b) Click play to run the collision. After the collision, allow the balls to move a few cm apart on the screen. Click on pause. Record your final momenta. Add them up. Record your answer below.

   c) Compare your answers for a) and b). What do you notice? (If your answers to a) and b) are not the same, then you have made a mistake. Go back and redo it.)

2. Let’s try to see if momentum is conserved when objects stick together. In the box on the right is a slider for elasticity. Set your elasticity to 0%.

   a) Where the play controls are, click on Restart. Determine your total initial momentum:

   b) Run the simulation. After the objects have collided, click on pause. Determine your total final momentum (click on each ball to make its data visible):

   c) Is momentum conserved when objects stick together?

1 a), b) Add the two momenta as signed numbers: a ball moving right has a positive momentum and a ball moving left has a negative one. Do the same after the collision.
1 c) The total initial momentum and the total final momentum are equal. Each ball’s momentum changes, but one ball gains exactly what the other loses.
2 a), b), c) With the elasticity at 0% the balls stick together and move off with one shared velocity. The total final momentum is again equal to the total initial momentum.

FINAL ANSWER

1 c) The totals in a) and b) are the same: total momentum is conserved

2 c) Yes, momentum is conserved when objects stick together

EXAMPLE

Dog and skateboard

A small dog of mass 6.5 kg runs at 12 m/s [E]. He jumps onto a 8.2 kg skate board that was moving towards him at a velocity of 2.3 m/s [W]. What is the final velocity of the dog and skateboard?

SOLUTION: Let East be + **Label the diagrams with all given information. Add an arrow to the "After" diagram to predict the direction they will be moving.** Use subscripts D and S for dog and Skateboard

The dog and skateboard are together at the end so they must have the same final velocity. Complete the solution

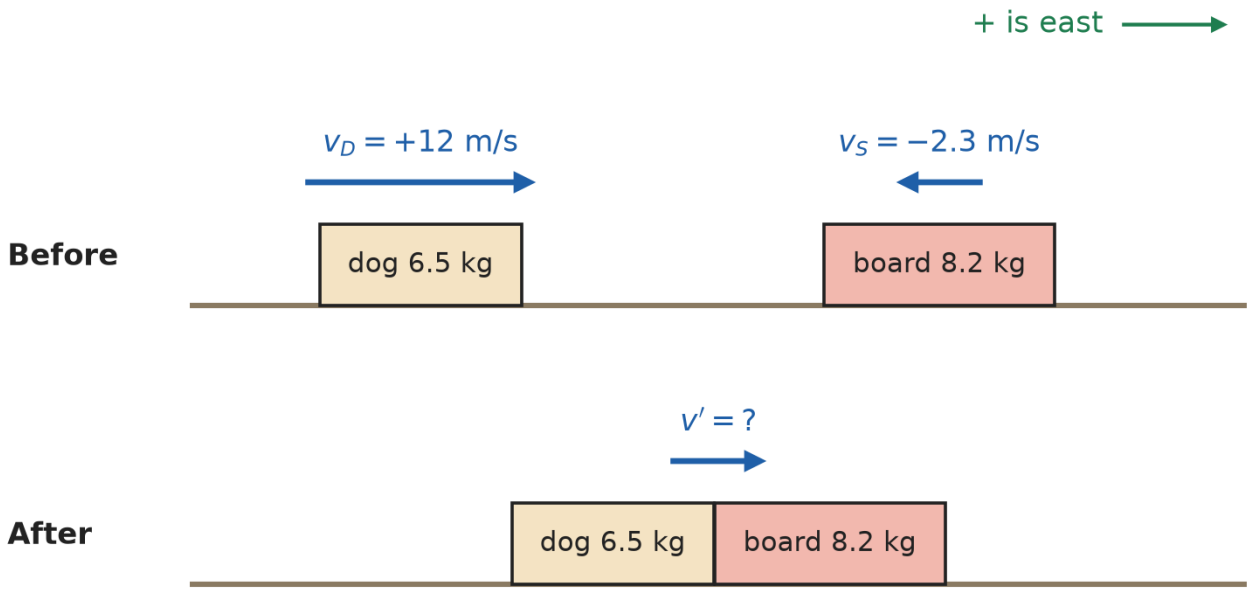
GIVEN

$m_D = 6.5$ kg $\quad v_D = +12$ m/s $\quad m_S = 8.2$ kg

$v_S = -2.3$ m/s $\quad v' = ?$

Sign convention: east is + (from the worksheet), so 2.3 m/s [W] is −2.3 m/s.

Before and after (+ is east)



Prediction: the dog has much more momentum than the skateboard, (6.5)(12) = 78 kg·m/s against (8.2)(2.3) = 18.86 kg·m/s, so together they move east.

They move together afterward, so $v_D' = v_S' = v'$. Isolate v' in letters first.

Formula: $m_D v_D + m_S v_S = m_D v_D' + m_S v_S'$

Put in $v_D' = v_S' = v'$: $m_D v_D + m_S v_S = (m_D + m_S)\, v'$

Divide both sides by $(m_D + m_S)$: $v' = \dfrac{m_D v_D + m_S v_S}{m_D + m_S}$

Put in the numbers:

$$v' = \frac{(6.5)(12) + (8.2)(-2.3)}{6.5 + 8.2} = \frac{78.00 - 18.86}{14.7} = +4.023 \text{ m/s}$$

v' is positive, so they move east, as predicted.

FINAL ANSWER

$v' = +4.0$ m/s

$\vec{v}' = 4.0$ m/s [E]

WHEN IS MOMENTUM CONSERVED?

From Newton’s third law to conservation of momentum

$m_2(\vec{v}_2' - \vec{v}_2) = -m_1(\vec{v}_1' - \vec{v}_1)$

Now, expand the brackets and rearrange to get the conservation of momentum formula

Expand both sides, then collect the “before” terms on the other.

Start: $m_2(\vec{v}_2' - \vec{v}_2) = -m_1(\vec{v}_1' - \vec{v}_1)$

Expand: $m_2\vec{v}_2' - m_2\vec{v}_2 = -m_1\vec{v}_1' + m_1\vec{v}_1$

Add $m_1\vec{v}_1'$ and $m_2\vec{v}_2$ to both sides: $m_1\vec{v}_1' + m_2\vec{v}_2' = m_1\vec{v}_1 + m_2\vec{v}_2$

Swap the sides: $m_1\vec{v}_1 + m_2\vec{v}_2 = m_1\vec{v}_1' + m_2\vec{v}_2'$

FINAL ANSWER

$m_1\vec{v}_1 + m_2\vec{v}_2 = m_1\vec{v}_1' + m_2\vec{v}_2'$

SUMMARY

Is the momentum of the system conserved?

In each of the situations below, determine of the momentum of the system would be conserved.

a) Two moving billiards balls bounce off each other.

b) A bouncy ball bounces off the floor. Its initial speed is equal to its final speed

c) A hockey puck rebounds off the boards

d) Two cars collide and stick together

e) A child rolling on a on a skateboard throws a ball forwards and slows down

Momentum is conserved when the only important forces are the equal and opposite forces the objects in the system put on each other ($\vec{F}_{net1} = -\vec{F}_{net2}$). If something outside the system pushes on it, the system’s momentum changes.

Situation	Conserved?	Why
a) Two billiard balls	Yes	The system is the two balls. During the collision the only large forces are the ones the balls put on each other.
b) Ball bouncing off the floor	No	The system is the ball alone. The floor is outside it and pushes up on the ball, reversing its momentum from $-mv$ to $+mv$. The speed is the same but the direction is not.
c) Puck off the boards	No	The system is the puck alone. The boards are outside it and exert a net force on the puck.
d) Two cars that stick	Yes	The system is both cars. Sticking together loses kinetic energy, not momentum.
e) Child throws a ball	Yes	The system is the child, skateboard and ball. The child pushes the ball forward and the ball pushes the child back equally, so the child slows down and the total stays the same.

FINAL ANSWER

a) Yes b) No c) No d) Yes e) Yes